

1. Phase Space

Phase space is an imaginary space in which every possible state of a system is represented by a point whose coordinates are the position and momentum of all particles.

2. Boson

Bosons are particles having integer spin (0, 1, 2, ...) and obey Bose–Einstein statistics.

Examples: Photon, Gluon

3. Fermion

Fermions are particles having half-integer spin (1/2, 3/2, ...) and obey Fermi–Dirac statistics.

Examples: Electron, Proton, Neutron

4. Internal Energy

Internal energy is the total energy possessed by a system due to the random motion and interaction of its molecules.

Unit: Joule (J)

5. Thermodynamic Potentials

Thermodynamic potentials are energy functions used to describe the equilibrium state of a thermodynamic system.

They are:

- Internal Energy (U)
- Enthalpy (H)
- Helmholtz Free Energy (F)
- Gibbs Free Energy (G)

7. Stefan's Constant

Stefan's constant is the proportionality constant used in the Stefan–Boltzmann law.

Value: $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

8. Gibbs Free Energy

Gibbs free energy is the energy available to do useful work at constant temperature and pressure.

Formula: $G = H - TS$

where, H = Enthalpy
T = Absolute Temperature
S = Entropy

9. Mean Free Path

Mean free path is the average distance travelled by a gas molecule between two successive collisions.

Symbol: λ

10. $C_p - C_v = R$: The difference between the molar specific heat at constant pressure (C_p) and constant volume (C_v) for an ideal gas is equal to the universal gas constant (R).

Formula: $C_p - C_v = R$

11. Heat Capacity

Heat capacity is the amount of heat required to raise the temperature of a body by 1 K (or 1°C).

Formula: $C = Q/\Delta T$

Unit: J K^{-1}

15. Fermi-Dirac Distribution.

The Fermi–Dirac distribution law gives the probability that an energy state is occupied by a fermion at a given temperature. It is applicable to particles having half-integer spin, such as electrons, protons, and neutrons. Since fermions obey the Pauli Exclusion Principle, no two identical fermions can occupy the same quantum state simultaneously.

Fermi–Dirac Distribution Function

The probability that a state of energy E is occupied by a fermion is given by

$$f(E) = \frac{1}{e^{(E-E_F)/kT} + 1}$$

where, $f(E)$ = Probability of occupation of an energy state, E = Energy of the state, E_F = Fermi energy, k = Boltzmann constant, T = Absolute temperature (K)

Characteristics

At absolute zero (0 K), all energy states below the Fermi energy are completely filled.

At 0 K, all states above the Fermi energy are empty.

At any temperature, $E = E_F$, $f(E) = \frac{1}{2}$.

As temperature increases, some electrons gain energy and move to higher energy levels.

The distribution is mainly used to describe the behavior of electrons in metals and semiconductors.

Graph:

At 0 K, the graph is a step function:

Below E_F : $f(E) = 1$, Above E_F : $f(E) = 0$

Applications: **1)** Explains the behavior of electrons in metals. **2)** Used in semiconductor physics. **3)** Helps explain electrical conductivity. **4)** Used in solid-state physics. **5)** Important in quantum mechanics and electronic devices.

Advantages: **1)** Correctly explains the distribution of electrons. **2)** Satisfies the Pauli Exclusion Principle. **3)** Accurately describes the properties of metals and semiconductors.

Limitations: **1)** Applicable only to fermions. **2)** Not suitable for bosons. **3)** It is a quantum statistical law and is not applicable to classical gases.

Conclusion

The Fermi–Dirac distribution describes how fermions are distributed among available energy states at thermal equilibrium. It is fundamental in quantum statistics and is widely used to explain the behavior of electrons in metals, semiconductors, and other solid materials.

Stefan–Boltzmann Law (10 Marks)

The total radiant energy emitted per unit surface area of a black body in one second is directly proportional to the fourth power of its absolute temperature.

Formula: $E = \sigma T^4$

or $P = \sigma AT^4$

where, E = Radiant energy emitted per unit area per second (W/m^2), P = Total power radiated (W), A = Surface area of the body (m^2), T = Absolute temperature (K), σ = Stefan–Boltzmann constant

Value of Stefan's constant:

$$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$$

Derivation: According to Planck's radiation law, the total energy emitted by a black body is obtained by integrating the energy over all wavelengths.

Hence, $E = \sigma T^4$

This shows that the energy emitted is directly proportional to the fourth power of the absolute temperature.

Therefore, $E \propto T^4$

Or, $E = \sigma T^4$

where σ is called the Stefan–Boltzmann constant.

Applications: **1)** To determine the energy emitted by stars and the Sun. **2)** To measure the surface temperature of stars. **3)** Used in furnaces and industrial heating. **4)** Used in thermal engineering. **5)** Used in infrared radiation and temperature measurement.

Limitations: **1)** Applicable only to black bodies. **2)** Real bodies require the emissivity factor. **3)** It gives only the total radiation and not the wavelength distribution.

Advantages: **1)** Helps calculate radiant energy. **2)** Useful in astronomy and astrophysics. **3)** Used in engineering and heat transfer calculations. **4)** Forms the basis of thermal radiation studies.

Conclusion: The Stefan–Boltzmann law is one of the fundamental laws of thermal radiation. It states that the energy radiated by a black body increases as the fourth power of its absolute temperature. The law is widely used in physics, astronomy, engineering, and thermal science.

Diagram:

